

MEDAN BELAWAN, Indonesia

WMO: 960330

Lat: 3.7883N

Lon: 98.7147E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	74.0	74.8	70.0	110.2	81.4	71.2	115.1	80.4	18.2	82.2	14.5	83.5	2.3	250	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.3	91.2	80.2	90.3	80.0	89.6	79.9	82.0	88.6	81.5	88.1	80.9	87.6	8.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.4	86.7	79.4	152.5	86.1	78.9	150.3	85.7	45.9	89.0	45.1	88.5	44.6	88.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	14.0	11.9	71.8	96.8	1.7	4.7	70.6	100.1	69.6	102.9	68.6	105.5	67.4	108.9		
(5)			69.6	84.4	1.7	1.3	68.4	85.3	67.4	86.0	66.4	86.7	65.2	87.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	81.3	82.0	83.1	83.5	83.4	83.4	82.9	82.4	81.9	81.5	81.5	81.2
	DBStd	1.77	1.60	1.41	1.54	1.64	1.62	1.67	1.61	1.65	1.56	1.43	1.27	1.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11807	971	897	1025	1004	1036	1002	1021	1006	958	975	945	967
	CDD65	6332	506	477	560	554	571	552	556	541	508	510	495	502
	CDH74	71693	5348	5335	6628	6758	6940	6711	6526	6183	5511	5394	5182	5177
	CDH80	25470	1580	1778	2553	2726	2797	2752	2530	2265	1873	1668	1504	1444
Wind	WSAvg	3.8	3.4	4.1	4.2	4.2	4.0	4.0	3.9	3.9	3.8	3.5	3.3	3.3
Precipitation	PrecAvg	100.5	6.3	3.7	5.8	5.6	8.4	6.9	7.2	9.3	12.1	13.3	10.9	11.0
	PrecMax	149.3	15.5	12.6	16.4	13.1	17.2	15.7	13.9	17.6	23.4	23.1	21.9	21.9
	PrecMin	65.8	2.8	1.4	1.2	1.1	3.5	3.1	2.2	3.7	6.5	6.5	3.7	3.7
	PrecStd	22.1	3.1	2.4	4.0	3.2	3.7	2.9	3.0	3.5	4.0	3.7	3.7	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	89.8	90.9	91.8	92.0	91.8	92.5	91.2	90.0	89.6	88.7	88.7
		MCWB	79.5	79.6	80.3	81.0	80.5	79.2	79.1	79.5	79.2	79.2	78.8	78.7
	2%	DB	87.9	88.6	89.9	90.5	90.5	90.4	90.0	89.7	88.9	88.1	87.5	87.5
		MCWB	78.8	79.0	80.0	80.6	80.5	79.8	79.8	79.9	79.4	78.9	78.8	78.6
	5%	DB	86.9	87.8	89.0	89.5	89.5	89.5	89.1	88.7	88.1	87.2	86.8	86.7
		MCWB	78.5	78.6	79.6	80.2	80.4	80.0	79.6	79.7	79.2	78.8	78.7	78.4
	10%	DB	86.0	86.9	88.1	88.6	88.7	88.6	88.2	87.7	87.2	86.3	86.0	85.7
		MCWB	78.3	78.4	79.2	79.8	80.1	79.7	79.4	79.3	79.0	78.6	78.6	78.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	80.8	81.8	82.4	82.7	82.4	82.7	84.3	82.0	80.7	80.1	80.4
		MCDB	86.8	87.4	88.7	89.4	89.6	88.7	89.6	88.2	87.6	86.5	85.9	86.4
	2%	WB	79.9	80.0	80.9	81.5	81.7	81.5	81.0	81.1	80.4	80.0	79.8	79.7
		MCDB	86.0	86.5	87.9	88.5	88.6	88.0	87.8	87.1	87.0	85.9	85.6	85.5
	5%	WB	79.3	79.5	80.2	80.8	81.0	80.8	80.5	80.3	79.8	79.4	79.2	79.1
		MCDB	85.2	85.9	87.2	87.8	87.8	87.5	87.2	86.8	86.4	85.4	85.0	84.7
	10%	WB	78.9	79.0	79.8	80.2	80.5	80.2	79.9	79.7	79.2	79.0	78.9	78.7
		MCDB	84.6	85.3	86.6	87.0	87.2	86.9	86.5	86.3	85.7	85.0	84.7	84.1
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	10.4	10.3	10.4	10.7	10.9	10.5	10.3	9.5	9.1	8.9
		MCDBR	10.4	10.7	11.2	11.1	11.3	11.4	12.1	11.6	11.7	10.7	10.1	10.2
		MCWBR	4.3	4.3	4.7	4.7	5.1	5.4	5.6	5.6	5.2	4.6	4.2	4.2
	5% WB	MCDBR	9.8	10.2	10.7	10.4	10.6	10.8	11.2	10.9	10.9	9.9	9.3	9.3
MCWBR		4.5	4.6	4.9	4.9	5.1	5.6	5.7	5.8	5.4	4.7	4.3	4.3	
Clear-Sky Solar Irradiance	taub	0.477	0.521	0.504	0.473	0.468	0.480	0.471	0.489	0.488	0.495	0.452	0.454	
	taud	2.265	2.117	2.184	2.280	2.313	2.279	2.308	2.241	2.246	2.220	2.352	2.347	
	Ebn at Noon	270	260	264	267	262	255	259	259	264	264	275	274	
	Edh at Noon	44	52	49	44	41	42	41	45	45	47	40	40	
All-Sky Solar Radiation	RadAvg	1405	1561	1625	1542	1514	1542	1501	1500	1437	1376	1297	1283	
	RadStd	84	86	91	71	88	89	79	106	114	151	59	90	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.34	+0.20	N/A	N/A	+0.72	+1.01	N/A	N/A	N/A	N/A

Nomenclature: See separate page